NZ HEALTH SURVEY ANALYSIS

End-to-End Data Pipeline & Power BI Portfolio Case Study

NZ Ministry of Health, NZ Health Survey 2024/25 | 2011–2025

Grace McNabb

Bachelor of Commerce (Information Systems)
University of Canterbury | 23 March 2026

1. Business Problem

New Zealand's health system faces significant and persistent challenges around chronic disease, mental health access, and health equity across population groups. The NZ Health Survey, conducted annually by the Ministry of Health, provides one of the most comprehensive longitudinal datasets on population health in New Zealand, tracking 180+ indicators across demographics, regions, and deprivation levels from 2011 to 2025.

This project addresses the following core question:

> *How have key health indicators changed across the New Zealand population from 2011 to 2025, and where do the largest health equity gaps exist across demographic and deprivation groups?*

The project demonstrates a complete end-to-end data analytics workflow: raw data ingestion, Python-based cleaning and transformation, Power BI data modelling, and a four-page interactive dashboard with a written findings report.

2. Data Source

Dataset Overview	
Source	New Zealand Ministry of Health, NZ Health Survey 2024/25
Access	Publicly available via the NZ Health Survey Annual Data Explorer
Coverage	2011 to 2025 (14 years of longitudinal data)
Files	Three CSV files: Prevalence/Mean, Changes Over Time, Subgroup Comparisons
Indicators	180+ health indicators across lifestyle, chronic disease, mental health, and healthcare access
Subgroups	Age group, ethnicity, gender, disability status, deprivation quintile, and health region

File Summary	
Prevalence/Mean (5,946KB)	Main dataset. Prevalences and means for total population and subgroups for 2024/25 and previous years. Used for executive summary, equity, and demographic analysis.
Changes Over Time (1,885KB)	Wide-format time series data with one column per survey year (2011–2024). Reshaped to long format via Python melt transformation. Used for trend analysis.
Subgroup Comparisons (329KB)	Adjusted rate ratios comparing population groups (e.g. Māori vs non-Māori) for 2024/25. Used for health equity analysis.

3. Python Data Pipeline

A five-script Python pipeline was developed to clean, transform, and validate all three source files before loading into Power BI. Scripts were written in VS Code with exploratory analysis conducted in a Jupyter Notebook.

Pipeline Scripts	
01_explore.ipynb	Jupyter Notebook for initial data exploration. Examined dataset shape, null value counts, data types, flag distributions, and unique indicator values. Documents the analytical decisions made prior to cleaning.
02_clean.py	Cleaned the prevalence file. Renamed columns to snake_case, handled publishing flags (blank/E/S), set suppressed values to null, standardised text fields, and dropped irrelevant columns. Output: prevalence_clean.csv.
03_transform.py	Applied feature engineering to the cleaned prevalence file. Created confidence interval width, gender gap, group_category, decade, and high_prevalence columns. Output: prevalence_final.csv (47,708 rows, 27 columns).
04_clean_others.py	Cleaned the time series and rate ratios files. Reshaped time series from wide to long format using pandas melt, extracting year from column names (percent_11 → 2011). Stripped embedded flag characters (e.g. '9.4 e') from prevalence values. Output: time_series_clean.csv (82,595 rows) and rate_ratios_clean.csv (3,198 rows).
05_validate.py	Validation script run iteratively throughout dashboard development to verify data integrity, check for unexpected nulls, and confirm filter outputs matched Power BI visuals.

3.1 Key Cleaning Decisions

- Suppressed values (flag = S) set to null rather than dropped, preserves the fact that data exists but was statistically unreliable, avoiding misleading gaps in trend charts
- Estimate values (flag = E) retained with a boolean is_estimate column, allows Power BI to optionally highlight uncertain estimates
- Wide-to-long reshape on time series file, 13-year columns (percent_11 through percent_24) melted into a single year/prevalence pair, reducing the file from 13 value columns to 1 and enabling proper time intelligence in Power BI
- Embedded flag characters stripped from prevalence values (e.g. '9.4 e' → 9.4), the time series file stored suppression flags in line with numeric values, causing Power BI to read the column as text rather than decimal
- group_category derived column created, categorises 71 unique group values into 6 meaningful categories (Age Group, Ethnicity, Gender, Deprivation, Health Region, Total Population) for cleaner Power BI slicer behaviour
- Confidence interval width renamed to margin_of_error in the rate ratios file for stakeholder readability

4. Data Model

A star schema was implemented in Power BI with three fact-style tables connected via a shared DimYear dimension table. The DimYear table was created using a DAX UNION to combine unique year values from both the Prevalence and TimeSeries tables, enabling a single year slicer to filter across all tables simultaneously.

Data Model Structure	
Prevalence	Main analysis table. 47,708 rows. One row per indicator/year/subgroup combination. Contains prevalence totals, male/female breakdowns, confidence intervals, and derived columns.
TimeSeries	Trend table. 82,595 rows. One row per indicator/group/year after wide-to-long reshape. Contains prevalence values, p-values, and significant_change flag.
RateRatios	Health equity table. 3,198 rows. 2024/25 snapshot only. Contains adjusted rate ratios and elevated_risk flag for population group comparisons.
DimYear	Dimension table created via DAX UNION of Prevalence[year] and TimeSeries[year]. Enables single slicer to filter both tables. Connected via one-to-many relationships.

5. DAX Measures

Key Measures	
Obesity Rate	CALCULATE(AVERAGE(Prevalence[prevalence_total]), indicator = "Obese", group = "Total", population = "adults"), dynamic, responds to year slicer
Smoking Rate	Same pattern filtered to "Current Smokers"
Hazardous Drinking Rate	Same pattern filtered to "Hazardous Drinking Pattern (Total Population)"
Mental Health Unmet Need	Same pattern filtered to "Unmet Need For Mental Health Care And Addictions Services"
Daily Vaping Rate	Same pattern filtered to "Daily Vaping"
Elevated Risk Count	CALCULATE(COUNTROWS(RateRatios), RateRatios[elevated_risk] = TRUE()), count of indicators where a group faces elevated risk
Avg Rate Ratio	AVERAGE(RateRatios[rate_ratio]), average disparity magnitude across all elevated risk indicators
Avg Prevalence	AVERAGE(TimeSeries[prevalence]), used in trend line chart, responds to indicator and year slicers

6. Dashboard Overview

The dashboard comprises four report pages designed to guide the viewer from high-level population health overview through to granular demographic and equity analysis. A shared DimYear slicer synchronises year filtering across pages.

Page 1, Executive Summary

Provides a high-level snapshot of key adult health indicators. The year slicer enables comparison across the full 2011–2025 period.

- Five KPI cards: Obesity Rate, Smoking Rate, Hazardous Drinking Rate, Mental Health Unmet Need, Daily Vaping Rate, all dynamic and responsive to the year slicer
- Bar chart: Top health risk indicators by prevalence for the selected year
- Table: Indicator prevalence with male, female, and gender gap columns

Page 2, NZ Health Trends Over Time

Explores how health indicators have changed across the 2011–2025 period using the reshaped TimeSeries dataset.

- Line chart: Prevalence trend for a user-selected indicator from 2011 to 2024
- Stacked bar chart: Biggest movers, indicators with the largest change since 2011, with chronic pain and hazardous drinking showing the most significant movement
- Table: Indicators with statistically significant change (p-value < 0.05)

Page 3, NZ Health Equity Analysis

A 2024/25 snapshot of health disparities across population groups using adjusted rate ratios. Rate ratios above 1.0 indicate elevated risk relative to the reference group.

- Bar chart: Rate ratios by comparison group with a constant reference line at 1.0
- Donut chart: Proportion of elevated risk indicators by comparison group, Māori vs non-Māori, Pacific vs non-Pacific, and disabled vs non-disabled share the highest burden
- KPI cards: Elevated Risk Count and Average Rate Ratio
- Table: Highest risk indicators sorted by rate ratio, most deprived vs least deprived women on daily smoking shows a rate ratio of 6.33

Page 4, Population Deep Dive

Allows granular exploration of any indicator across demographic dimensions simultaneously.

- Bar chart: Prevalence by age group, 55–64 age group shows highest obesity rates followed closely by 45–54
- Bar chart: Prevalence by ethnicity, Pacific populations show highest obesity prevalence; Māori show highest daily smoking rates
- Bar chart: Prevalence by deprivation quintile, Quintile 5 (most deprived) consistently shows highest prevalence across risk indicators
- Bar chart: Male vs female comparison, males 75+ show highest heart failure prevalence

7. Key Findings

The following insights were derived from analysis of NZ Health Survey data spanning 2011 to 2025:

31.9%	13.7%	18.8%	8.8%	7.7%
Avg Obesity Rate	Avg Smoking Rate	Hazardous Drinking	Mental Health Unmet	Daily Vaping

- Past-year drinkers represent the highest prevalence indicator in the dataset, indicating alcohol consumption remains near-universal among NZ adults and contextualising the hazardous drinking rate of 18.8%.
- Chronic pain and hazardous drinking show the most significant movement in the trends analysis, representing the biggest changes in health burden over the 2011–2025 period.
- The most deprived vs least deprived women show a daily smoking rate ratio of 6.33, the starkest equity gap in the dataset, indicating smoking is heavily concentrated in high deprivation communities among women.
- Obesity shows an elevated risk count of 20 across comparison groups with an average rate ratio of 1.56. Māori, Pacific, and disabled populations all face disproportionately high obesity rates relative to their reference groups.
- Pacific populations show the highest obesity prevalence by ethnicity; Māori show the highest daily smoking rates, consistent with longstanding health equity research in NZ.
- Quintile 5 (most deprived) consistently shows the highest prevalence across risk indicators including smoking, obesity, and hazardous drinking, confirming a clear deprivation gradient across NZ health outcomes.
- Males aged 75+ show the highest heart failure prevalence, consistent with the intersection of age and gender risk factors for cardiovascular disease.
- The 55–64 age group shows the highest obesity rate, followed closely by 45–54, suggesting middle-aged cohorts carry the highest current obesity burden in NZ.

8. Limitations

- The rate ratios file covers 2024/25 only, equity gap analysis is not possible with the current dataset. A production solution would source historical rate ratio data to track whether equity gaps are widening or narrowing over time.
- Survey data relies on self-reporting for many indicators (e.g. smoking, alcohol, physical activity) which may introduce social desirability bias, particularly for stigmatised behaviours.
- The NZ Health Survey uses a stratified random sample, small subgroup estimates (flagged S or E) carry wide confidence intervals and should be interpreted cautiously. These were retained as null in the cleaned dataset rather than dropped.
- The group_category classification in the Python pipeline uses keyword matching which may misclassify some group values. A production implementation would use a lookup table for precise categorisation.

- No geographic mapping was possible as the dataset provides health region labels only, not spatial coordinates. A production solution would join to a health region boundary shapefile for map visualisation.

9. Skills Demonstrated

Technical Skills	
Python	pandas (data cleaning, melt/reshape, feature engineering), Jupyter Notebook (EDA), VS Code (production scripts), iterative validation across pipeline stages
Data Cleaning	Flag handling, null management, embedded character stripping, wide-to-long reshaping, snake_case standardisation, derived column creation
Data Modelling	Star schema with three fact tables, DAX UNION for shared dimension table, cross-table year slicer synchronisation
DAX	CALCULATE with multiple filters, AVERAGE, COUNTROWS, dynamic KPI measures responsive to slicers, UNION for dimension tables
Power BI	Multi-page dashboard, synchronised slicers, conditional formatting, reference lines, KPI cards, bar/line/donut/table visuals
Health Analytics	Rate ratio interpretation, confidence interval analysis, deprivation quintile analysis, NZ health equity context

Ngā mihi, Grace McNabb | 23 March 2026